MambaMTL: Progressive Mutual-Guided Mamba Multitask Learning for Hyperspectral Image Pansharpening and Classification

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***Abstract*—Multitask learning (MTL) serves as an effective technology to improve both the performance of hyperspectral image (HSI) pansharpening and that of downstream classification tasks. However, most of the existing MTL frameworks overlook the close relationship between the two tasks, resulting in suboptimal performance. To solve the problem, we propose a progressive mutual-guided Mamba MTL framework (MambaMTL), which achieves mutual enhancement between the two tasks by interacting with the refined spatial–spectral features and the rich class semantic information generated by each task. Specifically, we design a dual-branch Mamba-Unet subnetwork with class-aware refinement (MTL-CRMNet) for HSI pansharpening. It incorporates parallel multihead class-aware refinement layers (MCRLs) to enhance object details in the reconstructed HSI by enriching class-specific features with semantic information from the classification network, thereby improving pansharpening performance. Besides, we introduce a Mamba-based pyramid subnetwork with hierarchical guided fusion (MTL-MGFNet) for HSI classification. It utilizes the pansharpened HSI to guide multiscale feature fusion in a feature pyramid manner, enhancing classification accuracy. Notably, both MTL-MGFNet and MTL-CRMNet share the same multiscale feature encoder, enabling the classification subnetwork to generate spatial–spectral features rich in texture and semantics. The interaction between tasks significantly enhances their respective performance. The evaluation results conducted on relevant datasets illustrate the superiority of the proposed MambaMTL in simultaneously achieving better HSI classification accuracy and higher reconstructed image quality.**

***Index Terms*—Hyperspectral classification, hyperspectral (HS) pansharpening, multitask learning (MTL), mutual guidance.**

Received 2 May 2025; revised 28 June 2025 and 19 August 2025; accepted 22 September 2025. Date of publication 29 September 2025; date of current version 9 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62201423, Grant 62471359, and Grant 62476013; in part by the Key Research and Development Program of Shaanxi under Grant 2025SF-YBXM-513; in part by the Young Talent Fund of Association for Science and Technology in Shaanxi under Grant 20230117 and Grant 20250133; and in part by the Fundamental Research Funds for the Central Universities under Grant QTZX25084. *(Corresponding author: Song Xiao.)*

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Digital Object Identifier 10.1109/TGRS.2025.3615140

I. Introduction

HYPERSPECTRAL image (HSI) pansharpening and classification are both popular and challenging tasks in HSI processing. They have been widely applied in image quality enhancement [1], land cover classification [2], and other related fields [3]. With the rapid development of deep learning techniques, both HSI pansharpening and classification have achieved remarkable progress. However, most existing studies treat these two tasks independently, without fully exploring the strong correlation between them. In fact, HSI pansharpening and classification are inherently complementary tasks, whose performance can be mutually reinforced through joint optimization. Therefore, it becomes a promising research direction to jointly learn these two tasks within a unified multitask learning (MTL) framework, which could potentially enable the two tasks to benefit from each other and further improve overall performance.

MTL frameworks emphasize leveraging complementary information between related tasks to enhance their performance. In recent years, there has been much MTL literature, such as super-resolution (SR) and classification [4], SR and segmentation [5], and detection and classification [6]. Taking the joint optimization of image quality enhancement and classification as an example, most of the existing Joint optimization methods primarily fall into two categories: 1) Unidirectional guidance approaches [7], where one task is utilized to enhance the performance of the other, and they fail to leverage the mutual benefits these tasks can offer each other, leading to suboptimal results; and 2) joint optimization methods, which consider the intrinsic connection between tasks by designing a shared feature learning module [8]. However, these approaches lack explicit task-specific considerations and mutual guidance between tasks. This leads to insufficient exploration of the inherent tight coupling and complementarity between tasks, ultimately leading to suboptimal multitask performance.

In addition to the task-specific issues discussed above, the existing backbones such as convolutional neural networks (CNNs) and Transformers exhibit inherent limitations. CNNs lack global contextual modeling capabilities due to the constraints of local receptive fields. Transformers, although capable of capturing long-range dependencies, suffer from quadratic computational complexity. In recent years, state space models (SSMs), exemplified by Mamba [9], have gained prominence in computer vision for

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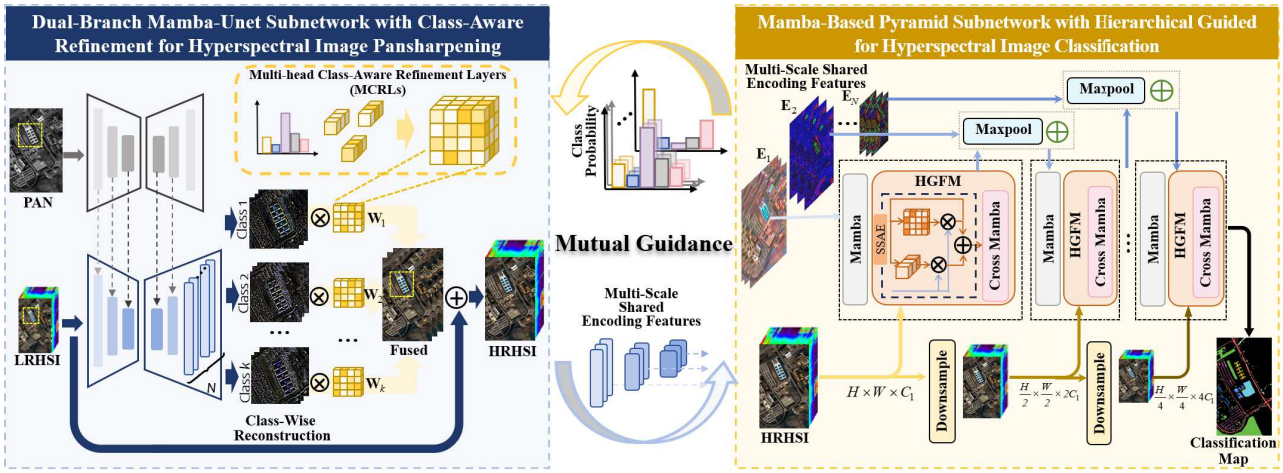


Fig. 1. Overview of the MambaMTL framework.

their ability to achieve linear-complexity global contextual modeling.

Based on the analysis above, as shown in Fig. 1, we propose a progressive mutual-guided Mamba MTL framework (MambaMTL), which cleverly integrates the HSI pansharpening and classification tasks within a unified framework, thereby boosting the performance of both tasks. Specifically, we design a dual-branch Mamba-Unet subnetwork with class-aware refinement (MTL-CRMNet) for HSI pansharpening, which introduces a set of paralleled multihead class-aware refinement layers (MCRLs). These layers refine the object details by enriching class-specific features with corresponding class semantic information generated by the classification network, thereby further improving the pansharpening performance. For HSI classification, we present a Mamba-based pyramid subnetwork with hierarchical guided fusion (MTL-MGFNet), which leverages the HSI reconstructed by the pansharpening network to guide multiscale features fusion in a feature pyramid manner, thereby assisting in improving the classification accuracy. It is worth noting that MTL-MGFNet and MTL-CRMNet share the same multiscale feature, enabling the classification subnetwork to produce multiscale spatial–spectral features that are rich in texture and semantics. Overall, the main contributions of this article are given as follows.

1) We introduce a progressive MambaMTL, which integrates the pansharpening and classification tasks within a unified framework. MambaMTL achieves mutual enhancement between the two tasks by interacting with the refined spatial–spectral features and the rich class semantic information generated by each task, thereby further improving their performance.
2) A dual-branch MTL-CRMNet is proposed for HSI pansharpening, which utilizes the pixel-level class semantic information generated by the classification subnet to augment the class-specific visual features, thereby further refining the object details in the reconstructed high-resolution hyperspectral image (HRHSI).
3) We present an MTL-MGFNet for HSI classification, which leverages the high-resolution features generated by the pansharpening subnetwork to guide multiscale feature fusion in a feature pyramid manner, thereby further improving classification accuracy.

The rest of this article is organized as follows. Section II presents the related work, Section III introduces the proposed MambaMTL method. In Section IV, we conduct a series of experiments to evaluate the effectiveness of the proposed MambaMTL. The conclusion is reported in Section V.

II. Related Work

A. HSI Pansharpening

HSI fusion techniques have garnered significant attention in the remote sensing community [10], [11], [12], as they overcome the limitations of single HSI SR by integrating complementary information from auxiliary images such as panchromatic (PAN), RGB, or multispectral images (MSIs) [13]. The HSI pansharpening task represents a typical application of HSI fusion, where PAN images serve as the auxiliary data to significantly enhance the spatial quality of the fused image while preserving spectral characteristics. The existing HSI pansharpening can be divided into two categories: 1) tradition approaches, including component substitution (CS) methods [14], multiresolution analysis (MRA) methods [15], Bayesian models [16], matrix factorization [17], and variational optimization [18]; and 2) deep learning-based methods, which achieve superior performance through supervised or unsupervised frameworks [19], [20], [21], [22], [23], [24], [25], [26], generating desirable fusion performance. For supervised HSI pansharpening, He et al. [19] first developed a specific network for hyperspectral (HS) pansharpening [hyperspectral pansharpening nerual network (HyperPNN)], which introduced a spectral predictive module to strengthen the spectral prediction capability of network. Subsequently, Dong et al. [20] presented a feature pyramid fusion network (FPFNet), which employs a dual-branch pyramid architecture to enable multiresolution feature extraction from PAN and HS inputs, progressively reconstructing HRHSI through coarse-to-fine feature fusion. Wang et al. [21] presented a multiscale pansharpening network (MDTP-Net) based on the dual Gaussian–Laplacian (DGLP)

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and Transformer. Lately, the diffusion model, as an advanced generative model, has also been applied to HSI pansharpening. He et al. [23] proposed a two-level semantic-driven diffusion (TSDD) method for HSI pansharpening. In the domain of unsupervised methods, Qu et al. [24] developed an interpretable unsupervised pansharpening network (IU2PNet) that successfully addresses the "black-box" architecture of conventional deep learning-based HSI pansharpening approaches while improving performance. Guarino et al. [25] introduced a novel unsupervised method that leverages a simple single-band pansharpening model integrated within a sequential band-wise adaptive framework, generating the satisfactory results. More recently, Shang et al. [26] presented a new unsupervised generative adversarial network framework (MFT-GAN), which employs a Transformer-based generator and two discriminators to effectively learn and fuse the spatial–spectral features from HSI and PAN images. More recently, zero-shot sharpening techniques, emerging as a novel form of unsupervised pansharpening, have attracted significant research attention [27], [28], [29], [30]. They not only mitigate the dependence on paired training data but also enhance robustness when transferring across different sensors. For example, Dian et al. [27] introduced a zero-shot learning method for HSI sharpening that quantitatively estimates the spectral and spatial responses from the observed HSI and MSI by alternating optimization, thereby obtaining the paired training samples and effectively addressing the challenge of limited training data availability. Guarino et al. [28] proposed a zero-shot HSI pansharpening method with hysteresis-based tuning, achieving uniform spectral quality across all bands through adaptive band-wise optimization. Particularly, Wang et al. [29] developed a universal zero-shot pansharpening network based on variational model, which extends spectral and spatial priors into a nonlinear space via zero-shot learning and employs variational iterations to achieve more precise solutions. Notably, it delivers satisfactory pansharpening performance even when applied across different sensors.

B. HSI Classification

In the early stage, some machine learning methods are used for HSI classification, including random forest (RF) [31], *k*-nearest neighbor (KNN) [32], and support vector machine (SVM) [33]. However, these approaches involved a lot of manual parameters, resulting in a poor classification performance. Recently, deep learning-based methods have proved that they can realize complex function approximation and effective feature representation of HSIs, including CNNs [34], [35], graph convolution networks (GCNs) [36], [37], Transformer networks (TNs) [38], and Mamba-based network [39], [40]. For example, Chhapariya et al. [35] introduced a spectral–spatial residual attention network (DSSpRAN) to extract the deep spectral and spatial features for classification. Recently, Mamba achieves higher computational efficiency than Transformers by integrating hardware awareness, leading to its application in HSI classification. Li et al. [39] designed a Mamba model-based (MambaHSI) framework composed of spatial and spectral Mamba blocks for HSI classification. Liu et al. [40] presented a new spectral–spatial adaptive Mamba for HSI classification, named as HyperMamba, improving the accuracy of classification. In addition, some joint classification methods have also attracted the attention of researchers. Zhu et al. [41] proposed a high- and low-frequency guided diffusion (HLF-GDiffusion) model for fusing PAN and MSI, thereby improving the classification performance.

C. MTL for HSI Pansharpening and HSI Classification

Currently, researchers have proposed numerous MTL networks for single image SR and classification tasks [7], [42], [43]. For example, Qu et al. [42] proposed a diffusion model-based MTL framework (MTLSC-Diff) for HSI SR and classification, which achieves the satisfactory results for both tasks. Sharma et al. [43] introduced a CNN-enabled network for joint SR and classification, which processes the low-resolution radar image through the SR network and then passes it for threat identification. However, research on MTL for HSI pansharpening and classification is relatively limited [8], [44]. Wu et al. [44] presented a novel classification-oriented MTL network (CMNet) for HSI pansharpening, which utilizes gradient guidance of the classification network to optimize the pansharpening network through multigradient learning.

D. Mamba

The concept of SSM was initially introduced in the S4 [45] model, which presents a unique architecture capable of effectively modeling global information. Since its introduction for linear-time sequence modeling in the NLP domain, as a variant of SSM, Mamba has quickly been varied to a wide applications in computer vision tasks. For instance, PanMamba [46] is a groundbreaking approach that utilizes Mamba to tackle the pansharpening challenge, achieving significant results even with simple Mamba blocks.

III. Proposed Method

A. Method Overview

A progressive MambaMTL consists of MTL-CRMNet and MTL-MGFNet and is presented to cooperatively improve the performance of HSI pansharpening and classification tasks in Fig. 2. MTL-CRMNet adopts the two similar Mamba U-shape structures to achieve hierarchical spatial–spectral features extraction and then reconstructs the HRHSI. In particular, a set of MCRLs are utilized to further augment the class-specific visual features using the specific class semantic information, thereby further refining the object details in the reconstructed HRHSI. MTL-MGFNet utilizes the HRHSI generated by MTL-CRMNet to guide shared multiscale spatial–spectral features fusion by the feature pyramid manner to produce fusion features rich in semantic information for classification, improving the classification accuracy. Therefore, MambaMTL boosts the performance of both tasks in HSI processing through the mutual enhancement of HS pansharpening and classification. The details of each part are shown as follows.

B. Dual-Branch MTL-CRMNet for HSI Pansharpening

As shown in Fig. 2, MTL-CRMNet consists of a dual-branch Unet stacked by Mamba blocks and FusionMamba blocks for HSI pansharpening and the paralleled MCRLs for HSI refinement.

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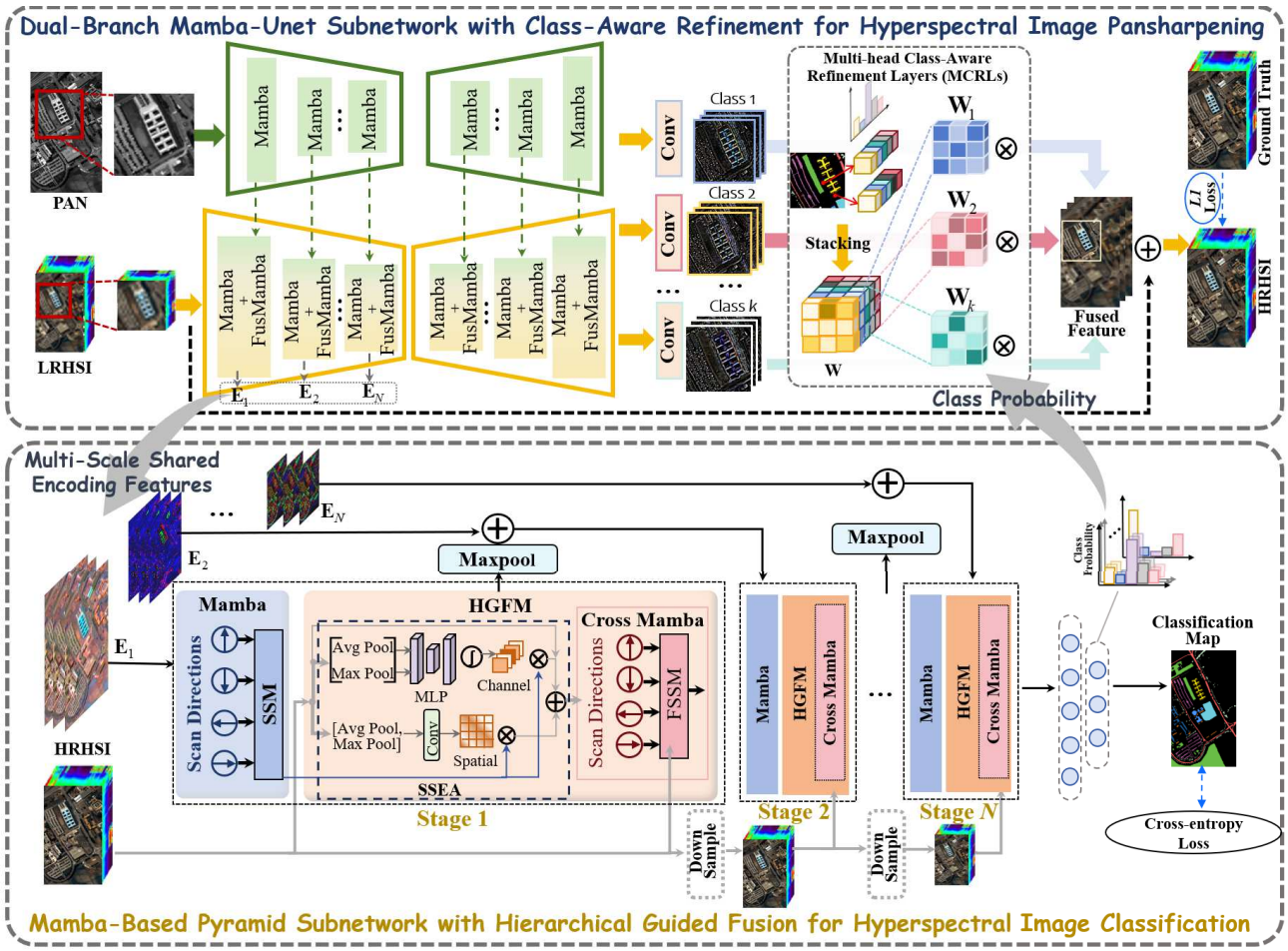


Fig. 2. Architecture of the proposed MambaMTL, which consists of two components: the HSI pansharpening subnetwork MTL-CRMNet for generating the desirable HRHSI and the HSI classification subnetwork MTL-MGFNet for producing the accurate land cover map.

1) Dual-Branch Mamba Unet for Multiscale Spatial–Spectral Feature Extraction and Fusion: Dual-branch Mamba-Unet consists of a spatial Unet and a spectral Unet. Spatial Unet is used for extracting multiscale spatial details from PAN image $\mathbf{P} \in \mathbb{R}^{H\times W\times 1}$, and the spectral Unet is used for capturing the multiscale spectral characteristic from low-resolution HSI (LRHSI) $\mathbf{H}_L$. In order to maintain the consistency between two Unet structures, the input $\mathbf{H}_L$ is upsampled by subpixel convolution to make it the same scale as the input $\mathbf{P}$, named as $\mathbf{H}_\uparrow \in \mathbb{R}^{H\times W\times C}$, where H and W denote the spatial dimensions and C represents the number of spectral bands.

To obtain sufficient deep spatial–spectral information without excessively increasing the number of subnetwork parameters, the proposed subnetwork MTL-CRMNet extracts features at N different scales (N is set to 3). At each scale, the spatial feature map and spectral feature map are processed through a Mamba block for feature extraction, respectively. Mamba blocks help model long-range dependencies within PAN and HS features. Subsequently, a fusion Mamba block is used for spatial–spectral information fusion. Specifically, $\mathbf{P}$ and $\mathbf{H}_\uparrow$ are input into a convolution layer and an activation function for initially shallow features encoding, respectively. Then, as for the PAN branch, each of the multiscale encoding layers owns a Mamba block and a downsample block for multiscale spatial–spectral features generation, followed by the initial feature extraction. Fig. 3 shows the details of the Mamba block. The spatial dimensions of $\mathbf{P}$ are halved with each layer of feature encoding. The formula of encoder of PAN branch is denoted by

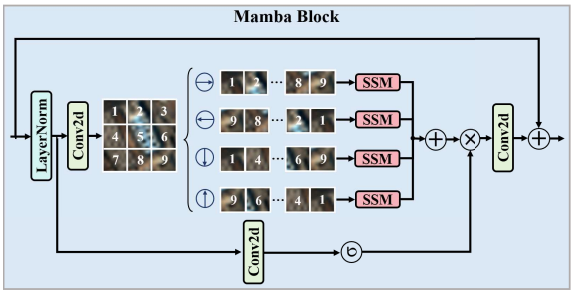


Fig. 3. Detailed structure of Mamba block.

$$\mathbf{P}_n^e = \begin{cases} \sigma_{\text{Leaky}}\left(f_I\left(\mathbf{P}\right)\right), & \text{n} = 1 \\ \Psi_\downarrow\left(f_M\left(\mathbf{P}_{n-1}^e\right)\right), & 2 \le \text{n} \le \text{N} \end{cases} \tag{1}$$

where $\mathbf{P}_n^e$ represents the encoded features of nth encoding layer; $f_M(\cdot)$ represents the Mamba block; $f_I(\cdot)$ and $\sigma_{\text{Leaky}}(\cdot)$ represent the initial convolution layer with 3×3 and the

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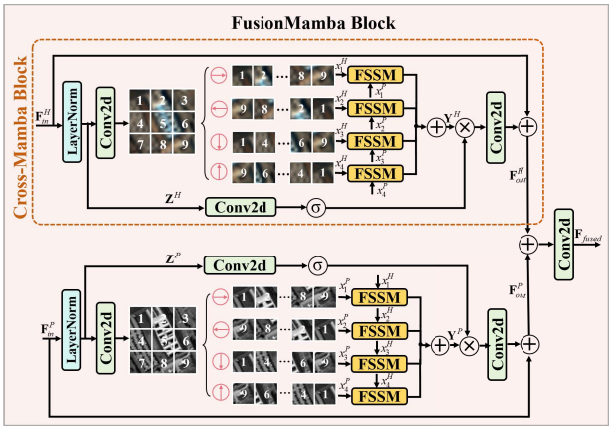


Fig. 4. Detailed structure of FusionMamba block.

LeakyReLU activation function, respectively; and $\Psi_{\downarrow}(\cdot)$ is the downsampling block composed of a convolution layer with scale size of 2×2 and a LeakyReLU activation function.

As for HS branch, the multiscale encoding features are generated by fusing the spatial–spectral features of $\mathbf{P}$ and $\mathbf{H}_{\uparrow}$ features at each scale based on the FusionMamba block. Then, the fused feature is downsampled by a downsampling block for the next multiscale features generation. The formula of encoder of HS branch is denoted by

$$\mathbf{H}_n^e = \begin{cases} \sigma_{\text{Leaky}}\left(f_I\left(\mathbf{H}_{\uparrow}\right)\right), & \text{n} = 1 \\ \Psi_{\downarrow}\left(f_{\Theta}\left(f_M\left(\mathbf{H}_{n-1}^e\right), \mathbf{P}_{n-1}^e\right)\right), & 2 \le \text{n} \le \text{N} \end{cases} \tag{2}$$

where $\mathbf{H}_n^e$ denotes the fused encoding features of nth encoding layer and $f_{\Theta}(\cdot)$ denotes the FusionMamba block.

As shown in Fig. 4, the FusionMamba block is used to associate and complement cross-modal features between PAN and HSI through feature fusion methods at various scales of the encoder–decoder layers. It extends Mamba to accommodate dual input, enabling the effective integration of different modalities. Specifically, the input features at each scale, denoted as $\mathbf{F}_{\text{in}}^P$ and $\mathbf{F}_{\text{in}}^H$, we obtain two sets of features

$$\mathbf{X}^m, \mathbf{Z}^m = f_x^m\left(\Phi_{\text{Norm}}\left(\mathbf{F}_{\text{in}}^m\right)\right), \quad f_z^m\left(\Phi_{\text{Norm}}\left(\mathbf{F}_{\text{in}}^m\right)\right), \ m \in \{P, H\} \tag{3}$$

where $\Phi_{\text{Norm}}(\cdot)$ represents layer normalization, $f_x^m(\cdot)$ and $f_z^m(\cdot)$ denote the two convolution layers, $\mathbf{F}_{\text{in}}^m$ is used as a unified representation for either the input PAN feature $\mathbf{F}_{\text{in}}^P$ or input HSI feature $\mathbf{F}_{\text{in}}^H$, and $\mathbf{X}^m$ and $\mathbf{Z}^m$ are the corresponding outputs. Subsequently, $\mathbf{X}^P$ and $\mathbf{X}^H$ are flattened individually in four directions to Cross-Mamba blocks for the purpose of information integration

$$\begin{cases} x_i^m = \Gamma_i\left(\mathbf{X}^m\right) \\ y_i^m = f_{\text{FSSM}}^{m,i}\left(x_i^m, x_i^{\widehat{m}}\right) \end{cases} \quad i = 1, 2, 3, 4. \tag{4}$$

Here, when m denotes H, $\widehat{m}$ is represented as P, and vice versa, $\Gamma_i(\cdot)$ is the flatten operation and $f_{\text{FSSM}}^m(\cdot)$ represents the fusion SSM (FSSM), which utilize the PAN and HS features as additional information for fusion, respectively. After that, the outputs are separately added to generate new feature map, denoted as $\mathbf{Y}^P \in \mathbb{R}^{H \times W \times C}$ and $\mathbf{Y}^H \in \mathbb{R}^{H \times W \times C}$. Similar to the calculation process of Mamba block, these features are subsequently combined to obtain the final fused feature map $\mathbf{F}_{\text{fused}}$, which can be formulated as

$$\begin{aligned} \mathbf{Y}^m &= \sum_{i=1}^{4} T_i\left(y_i^m\right) \\ \mathbf{F}_{\text{out}}^m &= f_F\left(\mathbf{Y}^m \cdot \sigma_{\text{SiLU}}\left(\mathbf{Z}^m\right)\right) + \mathbf{F}_{\text{in}}^m \\ &= f_{\text{CM}}\left(\mathbf{F}_{\text{in}}^m, \mathbf{F}_{\text{in}}^{\widehat{m}}\right) \\ \mathbf{F}_{\text{fused}} &= f_F\left(\mathbf{F}_{\text{out}}^P + \mathbf{F}_{\text{out}}^H\right) = f_{\Theta}\left(\mathbf{F}_{\text{in}}^P, \mathbf{F}_{\text{in}}^H\right) \end{aligned} \tag{5}$$

where $f_F(\cdot)$ denotes the 1×1 convolution layer, $\text{T}_i(\cdot)$ is the unflatten operation, $\sigma_{\text{SiLU}}(\cdot)$ is the SiLU activation function, $f_{\text{CM}}(\cdot)$ denotes the Cross-Mamba block, and $\mathbf{F}_{\text{fused}}$ represents the fused feature.

The decoder progressively reconstructs the encoding features and restores more details for reconstructing the desirable image. To capture contextual information, skip connections are also used in each stage of MTL-CRMNet to integrate the features from the encoder. At each decoding scale of PAN branch, these features are fused by a Mamba block and a upsampling block, which can be expressed as

$$\mathbf{P}_i^d = \begin{cases} \Psi_{\uparrow}\left(f_M\left(\mathbf{P}_N^e\right)\right) + \mathbf{P}_N^e, & i = 1 \\ \Psi_{\uparrow}\left(f_M\left(\mathbf{P}_{i-1}^d\right)\right) + \mathbf{P}_i^e, & 2 \le i < N \\ f_M\left(\mathbf{P}_{i-1}^d\right), & i = N \end{cases} \tag{6}$$

where $\mathbf{P}_i^d$ denotes the decoding feature of the ith layer at the PAN branch and $\Psi_{\uparrow}(\cdot)$ denotes the upsampled block composed of a deconvolution layer and a LeakyReLU activation function. Similarly, the decoding of HS branch can be written as

$$\mathbf{H}_i^d = \begin{cases} \Psi_{\uparrow}\left(f_{\Theta}\left(f_M\left(\mathbf{H}_N^e\right), \mathbf{P}_1^d\right)\right) + \mathbf{H}_N^e, & i = 1 \\ \Psi_{\uparrow}\left(f_{\Theta}\left(f_M\left(\mathbf{H}_{i-1}^d\right), \mathbf{P}_i^d\right)\right) + \mathbf{H}_i^e, & 2 \le i < N \\ f_{\Theta}\left(f_M\left(\mathbf{H}_{i-1}^d\right), \mathbf{P}_i^d\right), & i = N \end{cases} \tag{7}$$

where $\mathbf{H}_i^d$ denotes decoding feature of the ith layer at HS branch.

2) MCRLs for HSI Reconstruction and Refinement: Most of the existing pansharpening models typically adopt a one-fits-all approach for HRHSI reconstruction. In other words, these methods, these methods almost achieve the final high-resolution generation by sharing one or a few reconstruction layers. However, different land cover types exhibit distinct visual characteristics that should receive varying levels of attention during the reconstruction process. This raises the question: can we further improve the pansharpening performance by leveraging the class information? To address this, we introduce a set of paralleled MCRLs, which consist of multiple reconstruction blocks (RBs), where the number of RBs is equal to the number of land cover classes. Each RB, containing two convolution layers and a LeakyReLU activation function, is specifically designed to refine the texture details of a particular land cover class. Therefore, MCRLs could achieve the refinement of object details in the reconstructed HRHSI by pixel-level class-specific semantic information (class probabilities of reconstructed HRHSI). It can be expressed as follows:

$$\begin{aligned} \mathbf{W} &= f_{\text{cls}}\left(\mathbf{P}, \mathbf{H}_{\uparrow}\right) \\ \widehat{\mathbf{H}}_{\text{en}} &= \sum_{k=1}^{K} M_{\text{RB}}^k\left(\mathbf{H}_N^d\right) \cdot \mathbf{W}^k \end{aligned} \tag{8}$$

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where $f_{\text{cls}}(\cdot)$ denotes the pretrained classification network MTL-MGFNet, $\mathbf{W} \in \mathbb{R}^{H\times W\times K}$ is the pixel-level classification output of MTL-MGFNet with K classes, $\mathbf{W}^k \in \mathbb{R}^{H\times W\times 1}$ denotes the kth band of $\mathbf{W}$ corresponding to the class probability of the kth class, $\mathrm{M}_{\mathrm{RB}}^k(\cdot)$ is the kth RB, $\mathbf{H}_N^d$ is the output of the dual-branch Mamba-Unet given $\mathbf{P}$ and $\mathbf{H}_\uparrow$, and $\widehat{\mathbf{H}}_{\text{en}}$ denotes refined HRHSI. The weighted fusion ensures that even if MTL-MGFNet introduces some noise in the predicted labels, the reconstruction process remains robust by applying the appropriate RBs for feature enhancement.

The final HRHSI is obtained by adding $\mathbf{H}_\uparrow$ to the refinement reconstructed image, which can be summarized as

$$\widetilde{\mathbf{H}} = \widehat{\mathbf{H}}_{\text{en}} + \mathbf{H}_\uparrow \tag{9}$$

where $\widetilde{\mathbf{H}}$ denotes the final reconstructed HRHS image.

C. MTL-MGFNet for HSI Classification

As for HSI classification, we propose to establish the HSI classification based on image pyramid structure, which fully leverages the desirable HRHSI generated by MTL-CRMNet to guide the multiscale spatial–spectral features fusion for HSI classification. As shown in Fig. 2, MTL-MGFNet mainly consists of the dual-branch Mamba-based hierarchical guided fusion of high-resolution feature module, and the classifier with two fully connected layers and a RELU layer. In this section, we provide the details of the proposed MTL-MGFNet.

1) Mamba-Based Hierarchical Guided Fusion of High-Resolution Features for Classification: The shallow features of the encoder contain rich texture details and shape information. As the network depth increases, the encoding layers capture more global information, enabling the identification of overall shapes of land covers. Therefore, the multiscale encoding features in the pansharpening are shared for classification, thereby helping generating spatial–spectral features that are rich in texture and semantics. Besides, the HRHSI generated by the pansharpening subnetwork MTL-CRMNet is utilized in the form of feature pyramid to make up for the loss of spatial–spectral features during the encoding process, thereby constructing more distinguishable multiscale fused features. Similar to the scale of encoding layers in HSI pansharpening, the reconstructed HRHSI is downsampled into N scales, each downsampled half of the original image. At each scale, the generated HRHSI in the HSI pansharpening task guides the shared encoding feature to learn the latent spatial–spectral features for classification. Then, the multiscale fused feature is fed into the classifier to obtain the final HSI classification.

Taking the cropped pixelwise PAN and LRHS patch cubes $\mathbf{p} \in \mathbb{R}^{H\times W\times 1}$ and $\mathbf{h} \in \mathbb{R}^{H/4\times W/4\times C}$ as the input, the pretrained MTL-CRMNet first maps the cubes into four part: the reconstructed HRHSI, and shared multiscale encoding features, which can be formulated as

$$[\mathbf{H}, \mathbf{E}_1, \ldots, \mathbf{E}_N] = \Upsilon_{\text{HP}}(\mathbf{p}, \mathbf{h}) \tag{10}$$

where $\Upsilon_{\text{HP}}(\cdot)$ denotes the proposed pansharpening subnetwork; $\mathbf{H}$ represents the reconstructed HRHSI; $\mathbf{E}_1 \in \mathbb{R}^{H\times W\times C_1}$ represents the fused feature of first encoding layer of PAN and LRHS branches, which is the fused feature $\mathbf{P}_1^e$ and $\mathbf{H}_1^e$; and $\mathbf{E_N} \in \mathbb{R}^{H/(2^{N-1})\times W/(2^{N-1})\times 2^{N-1}C_1}$ represents the last encoding layer of PAN and LRHS branches, that is, $\mathbf{H}_3^e$. Subsequently, $\mathbf{H}$ and $\mathbf{E}_1$ are parallelly transmitted into the dual-branch Mamba-based hierarchical guided fusion of high-resolution feature module to supplement the missing detailed texture feature of shallow shared encoding spatial–spectral feature. Specifically, the shared encoding feature is first into the Mamba block to extract classification-relevant features, and a convolution layer with kernel size of 3×3 is utilized to extract the texture and structure feature of $\mathbf{H}$ for feature enhancement, which can be expressed as

$$\begin{aligned} \mathbf{F}_1 &= f_M(\mathbf{E}_1) \\ \mathbf{H}_1 &= f_{\text{conv}}(\mathbf{H}) \end{aligned} \tag{11}$$

where $f_{\text{conv}}(\cdot)$ is the convolution operation and $\mathbf{H}_1$ is the output of HRHSI $\mathbf{H}$ after shallow feature extraction.

Similar to the feature pyramid, the $\mathbf{H}_1$ is progressively decomposed into multiple scales. It allows to progressively transfer the hierarchical information from $\mathbf{H}_1$ to corresponding encoding features $\mathbf{E}_N$. The feature pyramid is constructed by progressively downsampling the high-resolution feature $\mathbf{H}_1$ to match the resolutions of $\mathbf{E}_N$, generating multiscale high-resolution features $\mathbf{H}_N \in \mathbb{R}^{H/(2^{N-1})\times W/(2^{N-1})\times 2^{N-1}C_1}$. At each level of the pyramid, the corresponding scale of $\mathbf{H}_N$ is fused with the $\mathbf{E}_N$ using the hierarchical guided fusion module (HGFM), which aims to selectively enhance the relevant details at each scale, ensuring that both fine-grained information from the high-resolution feature and the coarser contextual information from the shared encoding features are effectively preserved and integrated. It can be summarized as

$$\mathbf{Z}_i = \begin{cases} \Theta_M(\mathbf{H}_i, \mathbf{F}_i), & i = 1 \\ \Theta_M(\mathbf{H}_i, f_M(\mathbf{E}_i + f_\tau(\mathbf{Z}_i))), & 2 \le \mathrm{i} \le \mathrm{N} \end{cases} \tag{12}$$

where $\Theta_M(\cdot)$ denotes the HGFM function, $f_\tau(\cdot)$ represents the maxpooling operation for matching the multiscale encoding feature, and $\mathbf{Z}_i$ is the output of HGFM at the ith scale. Based on this, the Mamba-based hierarchical guided fusion of high-resolution features benefits from the multiscale contextual information, leading to a more comprehensive and discriminative feature representation for classification tasks.

Finally, $\mathbf{Z}_N$ is sent into the classifier to obtain the classification results, which can be written as

$$\mathbf{Y} = f_c(\text{Flatten}(\mathbf{Z}_N)) \tag{13}$$

where $f_c(\cdot)$ is the classifier, Flatten$(\cdot)$ is the flatten operation, and $\mathbf{Y}$ is the final classification result.

2) Hierarchical Guided Fusion Module: The HGFM operates in two steps: first, spatial–spectral joint enhancement attention (SSEA) is used to enhance the spatial–spectral representation of the encoding features by HRHSI, improving the spatial–spectral characterization capability. Second, the spatial–spectral-enhanced features are fed into a Cross-Mamba block to further compensate for the missing texture and structural information. The procedure can be written as

$$\mathbf{F}_e = f_{\text{SSEA}}\left(\overline{\mathbf{F}}, \overline{\mathbf{H}}\right) \tag{14}$$

where $f_{\text{SSEA}}(\cdot)$ represents the SSEA module, $\overline{\mathbf{F}}$ represents the shared encoding feature after Mamba block, $\overline{\mathbf{H}}$ represents the high-resolution guided information, and $\mathbf{F}_e$ is the enhanced

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feature. The specific process of enhancement guided by high-resolution features can be formulated as

$$\mathbf{F}_e = \Gamma_{\text{spa}}\left(\overline{\mathbf{H}}\right) \otimes \overline{\mathbf{F}} + \Gamma_{\text{spe}}\left(\overline{\mathbf{H}}\right) \otimes \overline{\mathbf{F}} + \overline{\mathbf{F}} \tag{15}$$

where $\Gamma_{\text{spa}}(\cdot)$ and $\Gamma_{\text{spe}}(\cdot)$ denote the spatial attention and channel attention, respectively. The spatial attention of SSEA module focuses on the most salient regions within the feature maps, ensuring that spatial details critical for classification are preserved. Concurrently, the spectral attention of SSEA emphasizes the most informative spectral bands, suppressing redundant or noisy information. This dual attention mechanism enhances the spatial–spectral feature representation, allowing the model to capture fine-grained details essential for discriminating between spectrally similar classes.

$\mathbf{F}_e$ is further sent into the Cross-Mamba block that compensates for missing details to aggregate multiscale contextual information and reinforces structural coherence by averaging cross-scale interactions and guided feature fusion. It ensures that both local fine details and global contextual dependencies are preserved, improving the accuracy of classification, which can be formulated as

$$\mathbf{Z} = f_{\text{CM}}\left(\mathbf{F}_e, \overline{\mathbf{H}}\right). \tag{16}$$

Overall, the optimization of the proposed MambaMTL is structured into three distinct stages. Stage 1 involves training the MTL-CRMNet subnetwork for initial pansharpening, where the LRHSI and its corresponding PAN image are input into the MTL-CRMNet subnetwork to extract and fuse spatial–spectral features. These decoded features are then processed through the MCRLs module to reconstruct the HRHSI, with weight coefficients set to 1/K. Finally, the upsampled LRHSI is combined with the reconstructed features via residual connections to produce the final HRHSI. Stage 2 focuses on training the MTL-MGFNet subnetwork for classification, utilizing the pretrained MTL-CRMNet model, where the LRHSI and PAN images are processed to generate HRHSI and multiscale shared features. These features are then input into the MTL-MGFNet subnetwork to extract classification-specific features, which are subsequently passed through a multilayer fully connected network to obtain the classification results. Stage 3 entails fine-tuning the MTL-CRMNet subnetwork based on the classification results, where all parameters, except those in the MCRL module, are frozen. The pixel-level classification results are utilized as weights for the MCRL module, and the subnetwork is fine-tuned using these adjusted weights to achieve the final refined HRHSI. The detailed algorithm is shown in Algorithm 1.

D. Loss Function

In this article, L1 loss is employed as the reconstruction loss to constrain the ideal HSI reconstruction because of its advantages in detail enhancement and resistance to noise interference compared to L2 loss, which can be expressed as

$$L_{\text{HP}} = \frac{1}{N_p}\sum_{i=1}^{N_p}\left\|\widetilde{\mathbf{H}}^{(i)} - \widehat{\mathbf{H}}^{(i)}\right\|_1 \tag{17}$$

Algorithm 1 Procedure of the Proposed MambaMTL Method

// Stage 1: Training of the MTL-CRMNet subnetwork.
Input: The PAN image $\mathbf{P} \in \mathbb{R}^{128\times128\times1}$, the LRHSI $\mathbf{H}_L \in \mathbb{R}^{32\times32\times C}$, and the reference HRHSI $\widehat{\mathbf{H}} \in \mathbb{R}^{128\times128\times C}$.
Output: The pansharpened HRHSI $\widetilde{\mathbf{H}} \in \mathbb{R}^{128\times128\times C}$.
for E epochs
$\mathbf{H}_L$ is interpolated to the same size as $\mathbf{P}$, named as $\mathbf{H}_\uparrow$.
Obtain the multi-scale encoding and decoding features of PAN image $\mathbf{P}$ by Eq. (1) and Eq. (3).
Obtain the multi-scale spatial–spectral fusion features of $\mathbf{P}$ and $\mathbf{H}_\uparrow$ by Eq. (2) and Eq. (4), the fusion strategy is based on Eq. (5) and Eq. (6).
Obtain the reconstructed HRHSI by Eq. (7), in which $w^k = \frac{1}{K} \in \mathbb{R}^{128\times128\times1}$.
Combine the reconstructed HRHSI and the upsampled LRHSI $\mathbf{H}_\uparrow$ to obtain the final HRHSI by Eq. (8).
Calculate the loss by Eq. (16), and optimize the parameters of MTL-CRMNet subnetwork by Adam optimizer.
end for
return $\widetilde{\mathbf{H}}$.
// Stage 2: Training of the MTL-MGFNet subnetwork.
Input: The PAN image $\mathbf{p} \in \mathbb{R}^{32\times32\times1}$, the LRHSI $\mathbf{h} \in \mathbb{R}^{32\times32\times C}$, the corresponding class label, and the pre-trained parameters of MTL-CRMNet subnetwork.
Output: The classification results $\mathbf{Y} \in {}^{1\times K}$ (Probability results belonging to each category).
Freeze the pre-trained parameters of MTL-CRMNet.
for F epochs
Generate the shared multi-scale spatial–spectral fusion encoding features and the final HRHSI by Eq. (9).
Extract the classification-relevant features of shallow shared fusion encoding feature, and the rich texture features of HRHSI by Eq. (10).
Fuse the shared multi-scale encoding features and corresponding HRHSI to obtain the classification features by Eq. (11), the fusion strategy is based on Eqs. (13)–(15).
Obtain the final classification result by Eq. (12).
Calculate the loss by Eq. (17), and optimize the parameters of MTL-MGFNet subnetwork by Adam optimizer.
end for
return $\mathbf{Y}$.
// Stage 3: Fine-tuning of the MTL-CRMNet subnetwork.
Input: The PAN image $\mathbf{P} \in \mathbb{R}^{128\times128\times1}$, the LRHSI $\mathbf{H}_L \in \mathbb{R}^{32\times32\times C}$, the reference HRHSI $\widehat{\mathbf{H}} \in \mathbb{R}^{128\times128\times C}$, and the corresponding pixel-level classification map $\mathbf{Y} \in \mathbb{R}^{128\times128\times K}$.
Output: The refined HRHSI $\widetilde{\mathbf{H}} \in \mathbb{R}^{128\times128\times C}$.
Freeze all parameters of the MTL-CRMNet subnetwork except MCRLs.
The training procedure is the same as that of **Stage 1**, except that Step 6 is replaced with: Obtain the reconstructed HRHSI by Eq. (7), in which $w^k = \mathbf{Y}_k$(kth band of $\mathbf{Y}$).
return $\widetilde{\mathbf{H}}$.

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TABLE I
DETAILED DESCRIPTION OF SIMULATED DATASETS

Dataset	Image Size	Band	Class	Pansharpening		Classification	
				Training	Testing	Training	Testing
PU	610 × 340	103	9	256	53	900	41876
PC	960 × 640	102	9	840	168	900	147252
HST	349 × 1905	144	15	535	107	750	14279
FR	990 × 990	63	9	64	225	900	179284

where N_p is the number of training samples for pansharpening and $\widehat{\mathbf{H}}^{(i)}$ represents the ith reference image.

As for HSI classification, we leverage the standard cross-entropy loss with label smoothing for the parameters of MTL-MGFNet learning, which measures the difference between the predicted class probabilities and the true class labels. This technique encourages the model to produce more confident but less overfitted predictions. It can be written as

$$L_{\mathrm{HC}} = -\sum_{i=1}^{K}\left((1-\varepsilon)\, y_i \log\left(p_i\right) + \frac{\varepsilon}{K}\left(1-y_i\right)\log\left(1-p_i\right)\right) \tag{18}$$

where K denotes the number of classes, y_i is the true label (1 for the correct class and 0 for others), p_i is the predicted probability for class i, and ε is the label smoothing parameter.

IV. EXPERIMENTAL RESULTS

A. Datasets

To verify the multitasks performance of the designed MambaMTL network, we employ the three simulated datasets (Pavia University (PU), Pavia Center (PC), and Houston (HST) datasets) and a real-world dataset (FR). As for simulated datasets, they generate the training and testing pairs by Wald's protocol for HSI pansharpening, where the spatial resolution of the PAN image is four times higher than that of the HSI. Specifically, the LRHSI is produced by downsampling the reference HRHSI with a 9 × 9 Gaussian filter. Meanwhile, the HR PAN image is obtained by averaging the visible bands of the reference HRHSI. For the HSI classification, we randomly selected a set of sample points of the labeled pixels. Specifically, the patch size is set to 32, and 100, 100, and 50 samples of each class are chosen for training, respectively.

The real-world FR dataset was acquired over the Florence area (Italy) by the PRISMA satellite, which could acquire both HSI and PAN images. The HSI contains 239 spectral bands with a spatial resolution of 30m. In the experiments, we used the remaining 63 pure bands after preprocessing. The corresponding PAN image has a spatial resolution of 5 m. The dimension of cropped HSI is 165 × 165 × 63, and the size of PAN image is 990 × 990. In the experiments, we intercepted small patches with the size of 20 × 20 and 120 × 120 of HSI and PAN images for training and testing. As for HSI classification, we have marked the area with nine common category labels based on Google Maps for classification evaluation, including roads, residential areas, railways, and grass. The cropped PAN patch size is set to 36, the corresponding LRHSI patch size is set to 6, and 100 samples of each class are chosen for training. Table I records the details of each dataset.

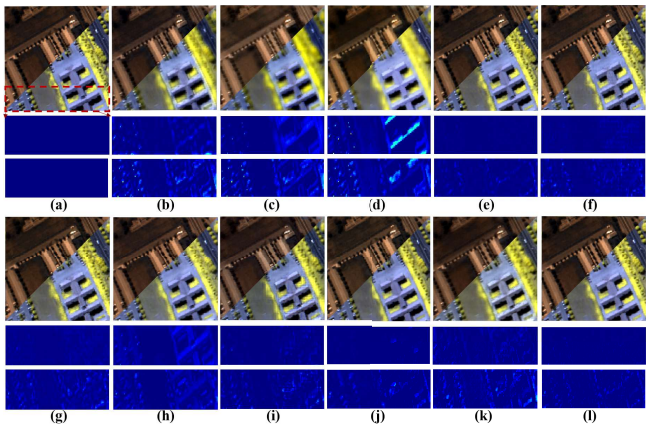


Fig. 5. First rows of each method: pansharpened images of the PU dataset (top-left region: bands 30–40–50 and bottom-right region: bands 70–80–90). Second and third rows of each method: corresponding visible and NIR-SWIR spatial error maps (blues means small differences, yellow means large differences, second rows: the mean absolute errors of 30–40–50 bands, and third rows: the mean absolute errors of 70–80–90 bands) of the competing methods for the PU dataset. (a) Reference. (b) SFIM. (c) GS. (d) CNMF. (e) SFIIN. (f) TransPAN. (g) SSAFF. (h) FPFNet. (i) PANMamba. (j) CMNet. (k) MTF-GAN. (l) MTL-CRMNet.

B. Experimental Settings and Quality Assessment

The proposed MambaMTL is implemented using the PyTorch framework and trained with two NVIDIA GeForce RTX 3090 GPUs. Specifically, MTL-CRMNet is trained for 400 epochs, and the Adam optimizer with an initial learning rate of 0.0005. The minibatch size is set to 4. MTL-MGFNet is trained for 200 epochs, and the learning rate is set to 0.0003. The training minibatch size is set to 128. Besides, when enhancing pansharpening performance using class information, we only fine-tune the class-aware components of the proposed MTL-CRMNet. During the fine-tuning phase, we set 200, 8, and 0.0003, as the training epochs, minibatch size, learning rate, and weight decay of Adam optimizer, respectively.

To verify the superiority of the proposed MambaMTL, ten pansharpening and eight classifications are used as competing methods. Specifically, the ten HSI pansharpening algorithms include SFIM [47], GS [48], CNMF [49], and seven published deep learning-based methods SFIIN [50], TransPAN [51], SSAFF [52], FPFNet [20], PANMamba [46], CMNet [44], and multiscale feature-guided transformer generative adversarial network (MFT-GAN) [26]. The eight deep learning-based classification algorithms include A2S2K [34], GMANet [53], SSFTT [54], HSIC-FM [55], MambaHSI [39], HSIMAE [56], HLF-GDiffusion [41], and CMNet [44].

For HSI pansharpening assessment, reconstructed images and error maps provide qualitative insights. On simulated datasets, quantitative evaluation is performed using $Q2^n$ [57], spectral angle mapper (SAM), and erreur relative globale adimensionnelle de synthese (ERGAS) to measure spatial, spectral, and overall quality. On a real-world dataset, the quality with no-reference (QNR) [58] index is employed, comprising spatial distortion (D_s) and spectral distortion (D_λ). Lower D_s values indicate better spatial fidelity, while values of (D_λ) closer to zero signify reduced spectral distortion. For HSI classification, qualitative assessment relies on visualiza-

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TABLE II
AVERAGE QUANTITATIVE ASSESSMENT FOR THE THREE SIMULATED DATASETS

Dataset	Pavia University			Pavia Center			Houston		
Metric	$Q2^n$	SAM	ERGAS	$Q2^n$	SAM	ERGAS	$Q2^n$	SAM	ERGAS
SFIM	0.8759	5.3707	4.3508	0.9071	5.4158	4.0644	0.8451	5.1765	4.1084
GS	0.8311	5.5267	5.1308	0.8574	5.8864	5.3591	0.7996	5.1090	4.7677
CNMF	0.8068	6.1150	5.8558	0.8442	6.0364	5.5035	0.8165	6.0448	5.0131
SFIIN	0.9628	3.2133	2.2829	0.9520	4.0781	2.5956	0.9163	3.8686	2.8556
TransPAN	0.9439	3.6692	2.6601	0.9497	4.4924	2.8508	0.9153	3.8363	3.0293
SSAFF	0.9412	3.7030	2.7540	0.9544	4.2052	2.7042	0.9355	3.3923	2.6714
FPFNet	0.9597	4.6107	2.8293	0.9586	4.0205	3.4855	0.9133	3.8995	3.7449
PANMamba	0.9633	4.0370	2.5758	0.9558	4.0504	3.1009	0.9310	3.6841	2.8096
CMNet	0.9546	4.8889	3.3222	0.9505	3.9039	2.7727	0.9264	3.7877	3.0026
MFT-GAN	0.9124	5.8405	3.9947	0.9171	5.2495	3.6789	0.8315	6.2463	4.4086
MTL-CRMNet	**0.9645**	**3.0609**	**2.1529**	**0.9628**	**3.8876**	**2.4689**	**0.9404**	**3.1796**	**2.4251**

TABLE III
QUANTITATIVE COMPARISON FOR THE REAL-WORLD FR DATASET

Method	SFIM	GS	CNMF	SFIIN	TransPAN	SSAFF	FPFNet	PANMamba	CMNet	MFT-GAN	MTL-CRMNet
QNR	0.5228	0.5615	0.6182	0.5705	0.7000	0.6952	0.6328	0.7209	0.7004	0.6484	**0.7297**
D_s	0.1926	0.1118	0.1259	0.1680	0.1006	0.0990	0.1251	0.0775	0.0837	0.1603	**0.0753**
D_λ	0.3525	0.3682	0.2943	0.3137	0.2216	0.2288	0.2781	0.2179	0.2362	0.2279	**0.2097**

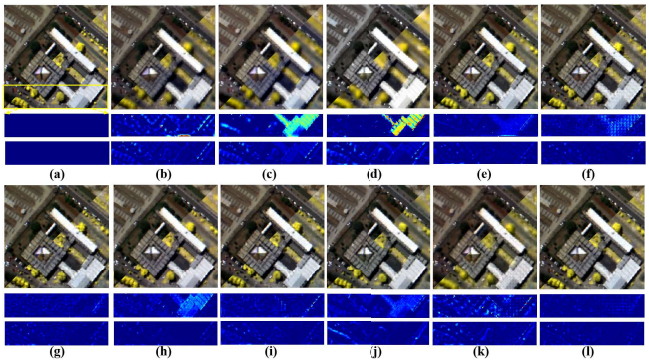


Fig. 6. First rows of each method: pansharpened images of the PC dataset (top-left region: bands 20–40–60 and bottom-right region: bands 70–85–100). Second and third rows of each method: corresponding visible and NIR-SWIR spatial error maps (blues means small differences, yellow means large differences, second rows: the mean absolute errors of 20–40–60 bands, and third rows: the mean absolute errors of 70–85–100 bands) of the competing methods for the PC dataset. (a) Reference. (b) SFIM. (c) GS. (d) CNMF. (e) SFIIN. (f) TransPAN. (g) SSAFF. (h) FPFNet. (i) PANMamba. (j) CMNet. (k) MTF-GAN. (l) MTL-CRMNet.

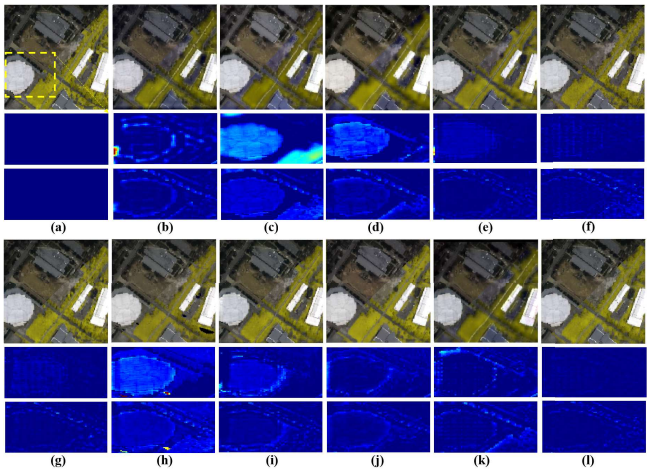


Fig. 7. First rows of each method: pansharpened images of the HST dataset (top-left region: bands 30–50–70 and bottom-right region: bands 70–90–100). Second and third rows of each method: corresponding visible and NIR-SWIR spatial error maps (blues means small differences, yellow means large differences, second rows: the mean absolute errors of 30–50–70 bands, and third rows: the mean absolute errors of 70–90–100 bands) of the competing methods for the HST dataset. (a) Reference. (b) SFIM. (c) GS. (d) CNMF. (e) SFIIN. (f) TransPAN. (g) SSAFF. (h) FPFNet. (i) PANMamba. (j) CMNet. (k) MTF-GAN. (l) MTL-CRMNet.

tion of classification maps, while quantitative performance is measured via overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa).

C. HSI Pansharpening Experimental Results

The comparative experimental results across four distinct scenarios are presented in Figs. 5–8, with corresponding quantitative metrics detailed in Tables II and III. It can be observed that the traditional methods, SFIM, GS, and CNMF, exhibit varying degrees of spatial and spectral distortion compared to deep learning-based approaches. These conventional techniques struggle to generate fine spatial details while maintaining spectral integrity, as evidenced by their higher SAM and RMSE values. Among deep learning-based methods, MSI pansharpening approaches such as SFIIN, TransPAN, SSAFF, and PANMamba demonstrate relatively lower PSNR values on the three simulated datasets (PU, PC, and HST datasets). The performance gap likely stems from their insufficient integration of spatially and spectrally discriminative features during fusion. Additionally, evident checkerboard artifacts produced by TransPAN, SSAFF, and PANMamba on the real-world FR dataset further validate the limitation. Furthermore, HSI pansharpening methods, FPFNet and CMNet, also exhibit suboptimal spectral fidelity, primarily due to their inadequate modeling of spectral continuity constraints within their network architectures. The unsupervised deep learning-based HS pansharpening method MTF-GAN demonstrates over-

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PU DATASET (BOLD INDICATES THE BEST RESULT FOR EACH LINE)

Class	A2S2K	GMANet	SSFTT	HSIC-FM	MambaHSI	HSIMAE	HLF-GDiffusion	CMNet	MTL-MGFNet
1	96.37	99.38	91.64	96.45	97.53	96.26	97.53	91.43	**99.56**
2	94.64	97.10	97.58	96.48	98.39	98.83	98.72	96.50	**99.28**
3	97.62	99.81	98.09	96.45	98.97	91.60	99.67	97.33	**100.00**
4	92.53	**99.74**	88.02	98.99	98.63	96.63	97.00	97.72	99.02
5	98.59	**100.00**	97.54	**100.00**	**100.00**	99.92	99.63	**100.00**	**100.00**
6	97.93	98.89	99.88	99.82	98.79	99.72	98.49	99.40	**100.00**
7	99.32	**100.00**	99.62	**100.00**	99.92	**100.00**	98.27	99.92	**100.00**
8	98.56	**99.78**	99.55	98.19	98.44	88.89	97.42	97.88	99.00
9	93.35	**100.00**	89.12	**100.00**	99.12	97.05	99.47	97.15	99.37
OA (%)	95.87	98.46	96.12	97.48	98.46	97.19	98.32	96.53	**99.45**
AA (%)	96.54	99.41	95.45	98.49	98.87	96.55	98.45	97.48	**99.58**
Kappa (%)	94.59	97.97	94.92	97.09	96.68	98.50	97.79	95.44	**99.27**

TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PC DATASET (BOLD INDICATES THE BEST RESULT FOR EACH LINE)

Class	A2S2K	GMANet	SSFTT	HSIC-FM	MambaHSI	HSIMAE	HLF-GDiffusion	CMNet	MTL-MGFNet
1	98.26	**100.00**	99.96	99.94	**100.00**	**100.00**	99.32	99.39	99.96
2	96.64	95.17	93.85	91.31	95.88	**98.17**	97.89	91.39	96.30
3	89.51	87.25	95.54	**98.55**	95.55	87.57	89.69	85.86	93.04
4	99.93	96.20	**100.00**	99.26	99.56	99.92	99.93	99.48	**100.00**
5	99.30	99.56	98.23	98.89	96.88	99.75	98.69	98.54	**99.80**
6	95.35	**99.58**	96.03	97.89	98.67	98.80	95.58	95.34	98.02
7	98.94	90.05	**99.15**	96.76	95.28	94.03	99.12	97.02	**99.15**
8	94.66	98.52	98.28	98.74	99.03	**99.51**	97.99	98.31	99.35
9	95.49	98.95	83.09	**100.00**	99.96	97.54	95.77	93.08	96.33
OA (%)	96.83	98.43	96.38	98.78	98.96	99.09	98.34	97.86	**99.21**
AA (%)	96.45	96.14	96.02	97.93	97.87	97.25	97.11	95.38	**98.00**
Kappa (%)	95.55	97.78	97.71	98.27	98.50	98.72	97.65	96.97	**98.89**

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HST DATASET (BOLD INDICATES THE BEST RESULT FOR EACH LINE)

Class	A2S2K	GMANet	SSFTT	HSIC-FM	MambaHSI	HSIMAE	HLF-GDffusion	CMNet	MTL-MGFNet
1	96.40	97.60	95.92	97.92	89.66	**100.00**	93.76	94.88	92.41
2	98.25	99.04	92.02	92.94	**99.48**	98.42	95.69	96.33	99.28
3	97.27	**100.00**	98.71	99.85	**100.00**	99.85	99.00	96.27	99.71
4	88.75	95.58	91.48	94.30	93.44	98.07	90.68	93.33	**98.31**
5	99.28	98.15	99.84	96.64	**100.00**	**100.00**	99.92	99.84	**100.00**
6	**100.00**	98.15	98.15	**100.00**	97.78	**100.00**	**100.00**	96.62	**100.00**
7	93.85	93.85	87.93	89.41	94.43	95.32	91.17	**95.50**	93.45
8	86.25	76.45	94.29	81.07	86.98	84.67	**94.86**	82.07	92.12
9	90.42	95.13	90.58	91.18	88.11	90.27	86.98	91.93	**97.68**
10	**100.00**	95.11	98.70	86.83	98.94	98.39	98.86	94.70	**100.00**
11	99.27	99.11	96.36	91.48	98.33	95.36	**100.00**	98.95	99.76
12	97.00	97.08	**100.00**	83.94	96.82	83.77	96.92	95.21	97.16
13	99.57	95.74	**100.00**	98.33	90.51	96.90	95.74	95.74	95.31
14	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	98.36	**100.00**
15	**100.00**	**100.00**	96.52	**100.00**	99.10	**100.00**	99.85	99.24	**100.00**
OA (%)	95.66	95.44	95.34	92.10	95.10	95.24	95.56	94.80	**97.09**
AA (%)	96.42	96.07	96.03	93.60	95.57	96.07	96.23	95.27	**97.47**
Kappa (%)	95.31	95.08	94.97	91.41	96.29	94.86	95.20	94.37	**96.86**

all superior performance compared to traditional algorithms, further validating the potential of unsupervised learning in the field of HS pansharpening. Nevertheless, due to the inherent limitations of reference-free learning, the fusion performance still lags behind that of supervised deep learning-based pansharpening approaches. Overall, the proposed MTL-CRMNet achieves state-of-the-art performance across all evaluation metrics ($Q2^n$, SAM, and ERGAS), striking an optimal balance between spatial resolution and spectral fidelity. Notably, its performance on the real-world FR dataset highlights the effectiveness of the proposed approach in enhancing the quality of LRHS images.

D. HSI Classification Experimental Results

Figs. 9–12 visually display the comparative classification results across four distinct scenarios, while Tables IV–VII provide the corresponding quantitative evaluation metrics. It can be observed that the results of A2S2K and SSFTT

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TABLE VII

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE FR DATASET (BOLD INDICATES THE BEST RESULT FOR EACH LINE)

Class	A2S2K	GMANet	SSFTT	HSIC-FM	MambaHSI	HSIMAE	HLF-GDiffusion	CMNet	MTL-MGFNet
1	99.71	99.43	**99.77**	93.08	97.62	98.92	99.22	99.07	98.39
2	87.64	87.44	80.76	87.17	**90.79**	83.93	86.34	85.52	88.62
3	65.97	72.84	84.85	81.03	77.10	73.89	**85.47**	75.81	81.63
4	**98.26**	92.56	95.66	92.62	92.31	93.89	96.82	93.58	96.18
5	68.65	**88.47**	77.72	73.51	72.51	66.13	80.97	80.14	84.62
6	96.28	97.74	97.61	95.22	94.40	96.26	96.11	**97.92**	97.62
7	99.32	98.62	**99.99**	98.42	97.82	98.40	98.52	98.29	98.88
8	94.61	93.21	90.57	**97.48**	93.70	94.27	88.12	91.89	94.34
9	99.86	99.86	93.63	98.78	88.44	94.86	99.01	**100.00**	99.50
OA (%)	87.14	89.94	87.11	88.20	88.70	85.57	89.26	88.28	**90.92**
AA (%)	90.03	92.24	91.17	90.81	89.41	88.93	92.26	91.36	**93.31**
Kappa (%)	84.22	87.67	84.36	85.59	85.00	82.45	86.88	85.67	**88.89**

TABLE VIII

FEWER SAMPLE CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE PU DATASET (BOLD INDICATES THE BEST RESULT FOR EACH LINE)

Index	Samples	A2S2K	GMANet	SSFTT	HSIC-FM	MambaHSI	HSIMAE	HLF-GDiffusion	CMNet	MTL-MGFNet
OA(%)	2	47.24	40.46	47.90	53.30	51.03	45.99	53.65	49.25	**55.30**
	5	69.74	70.24	75.48	75.80	64.11	61.84	71.98	64.12	**78.89**
	10	76.13	80.07	81.73	84.48	72.74	76.41	85.48	73.49	**87.80**
AA(%)	2	48.77	50.28	47.42	45.46	32.25	49.87	46.88	31.72	**51.72**
	5	71.87	81.38	66.26	80.25	76.22	67.81	71.12	66.48	**81.97**
	10	78.98	81.48	**91.49**	78.40	89.64	82.80	82.91	78.37	84.33
Kappa(%)	2	34.71	40.22	30.01	37.92	40.60	25.25	35.34	31.37	**43.19**
	5	62.69	64.75	63.26	67.61	69.32	71.25	53.46	55.04	**73.06**
	10	70.17	80.79	75.32	76.29	80.25	80.75	70.52	67.03	**83.52**

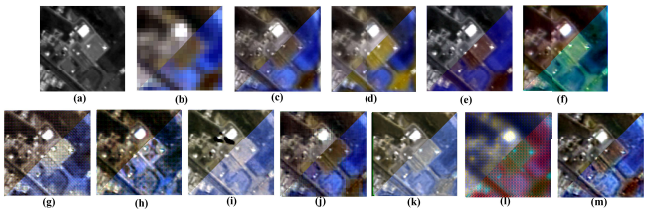


Fig. 8. Pansharpening results of the FR dataset with different methods (top-left region: bands 5–10–15 and bottom-right region: bands 40–50–60). (a) Real PAN image. (b) Real LRHS image. (c) SFIM. (d) GS. (e) CNMF. (f) SFIIN. (g) TransPAN. (h) SSAFF. (i) FPFNet. (j) PANMamba. (k) CMNet. (l) MTF-GAN. (m) MTL-CRMNet.

are relatively low on all four scenarios, showing the poor robustness of these methods. GMANet performs better for the real-world FR dataset, but still suffers from serious false detection. There exist many missed detection phenomena in the result of HSIC-FM, especially on the HST dataset. The accuracy of the results generated by MambaHSI and HSIMAE is comparable on three datasets of airborne (PU, PC, and HST datasets), while MambaHSI outperforms HSIMAE in terms of OA, AA, and Kappa indicators on real satellite dataset (FR dataset). The joint classification method, HLF-GDiffusion, achieves superior overall performance across four scenarios compared to CMNet, with this advantage primarily stemming from its hierarchical frequency-guided architecture that enables optimal cross-modal feature fusion. The proposed MTL-MGFNet outperforms the state-of-the-art methods with less miss detection and false detection, indicating the effectiveness of the proposed method in multimodal joint classification.

E. Ablation Study

1) Influence of Classification Performance on Fewer Training Samples: In order to verify the classification performance of the proposed MTL-MGFNet on fewer samples, we conducted the experiments on the PU dataset using a reduced number of training samples per class (specifically, 2, 5, and 10 pixels). The results listed in Table VIII demonstrate that our proposed method maintains superior performance across various metrics, even with significantly fewer training samples.

2) Contribution of Pansharpening Task to Classification Performance: To validate the influence of HSI pansharpening on classification performance, we conducted experiments on three datasets to evaluate the impact of training the multiscale feature encoder from scratch (abbreviated as "TFS") on classification performance, i.e., excluding the transferring of high-resolution features and multiscale shared features of HSI pansharpening subnetwork. As shown in Fig. 13, the results indicate a noticeable decrease in classification performance across three evaluation metrics, especially in OA and Kappa, when the classification subnetwork is trained from scratch. The results demonstrate the effectiveness of using a shared multiscale spatial–spectral feature encoder.

3) Contribution of Classification Task to Pansharpening Performance: MCRLs are utilized for the HRHSI refinement based on the class information. Therefore, we divide our MTL-CRMNet into two variations, i.e., model with MCRLs, and model without MCRLs (abbreviated as "w/o MCRLs"). In w/o MCRLs, the weight parameters of MCRLs are set to $1/k$, which indicates that the class semantic information does not participate in the pansharpening process. By comparing the performance of "MCRLs" and "w/o MCRLs," we

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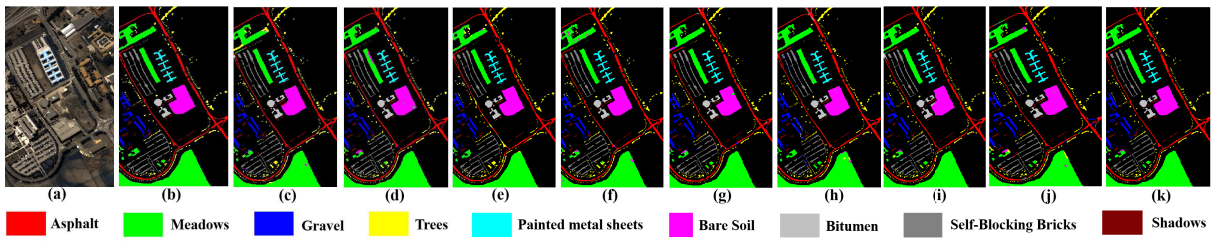


Fig. 9. Classification maps of different methods on the PU dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) A2S2K. (d) GMANet. (e) SSFTT. (f) HSIC-FM. (g) MambaHSI. (h) HSIMAE. (i) HLF-GDiffusion. (j) CMNet. (k) MTL-MGFNet.

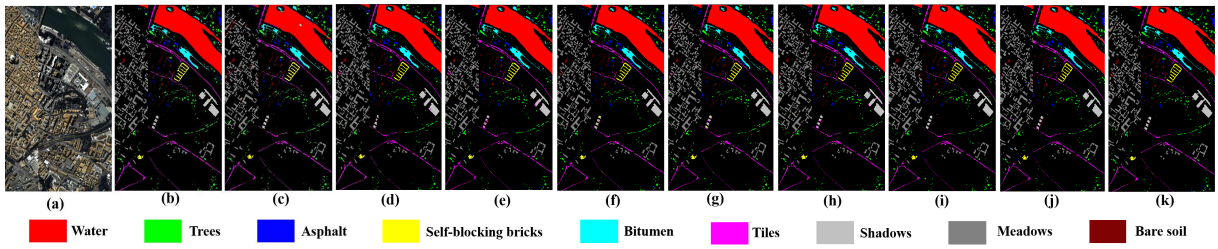


Fig. 10. Classification maps of different methods on the PC dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) A2S2K. (d) GMANet. (e) SSFTT. (f) HSIC-FM. (g) MambaHSI. (h) HSIMAE. (i) HLF-GDiffusion. (j) CMNet. (k) MTL-MGFNet.

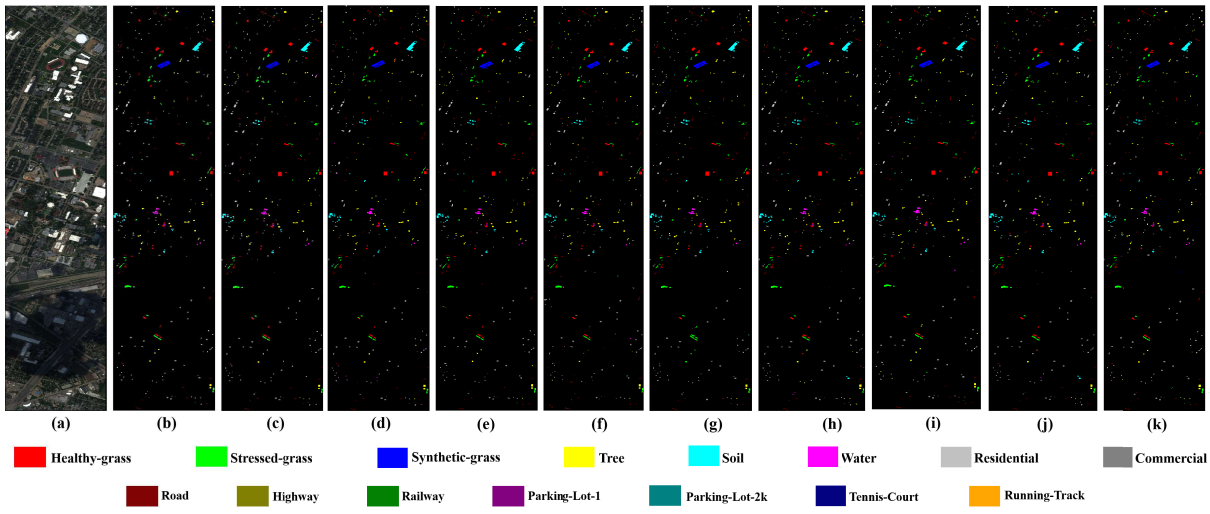


Fig. 11. Classification maps of different methods on the HST dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) A2S2K. (d) GMANet. (e) SSFTT. (f) HSIC-FM. (g) MambaHSI. (h) HSIMAE. (i) HLF-GDiffusion. (j) CMNet. (k) MTL-MGFNet.

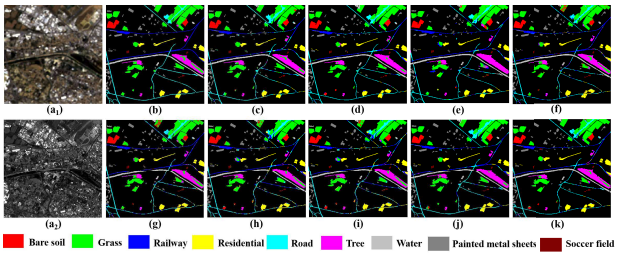


Fig. 12. Classification maps of different methods on the FR dataset. (a1) Real LRHS image. (a2) Real PAN image. (b) Ground-truth map. (c) A2S2K. (d) GMANet. (e) SSFTT. (f) HSIC-FM. (g) MambaHSI. (h) HSIMAE. (i) HLF-GDiffusion. (j) CMNet. (k) MTL-MGFNet.

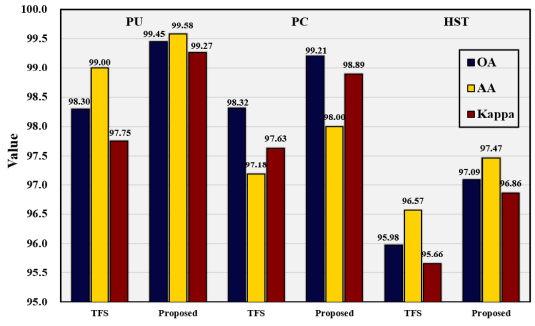


Fig. 13. Quantitative metrics comparison results for the contribution of pansharpening task to classification performance.

validate the impact of the classification component on fusion performance. As shown in Fig. 14, model "MCRLs" obtains the best performance, especially in the SAM, which indicates the effectiveness of the proposed MTL-CRMNet. In addition,

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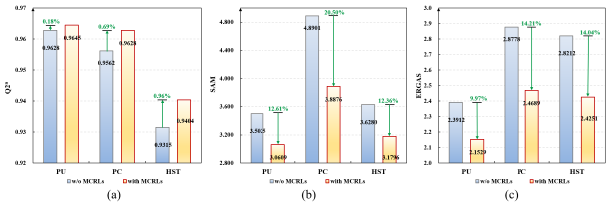


Fig. 14. Quantitative metrics comparison results for the contribution of classification task to pansharpening performance.

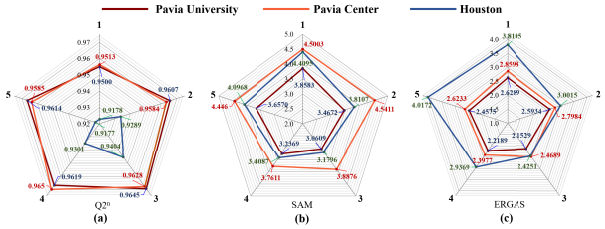


Fig. 15. Comparative analysis of pansharpening performance for MTL-CRMNet with different scale parameters. (a) $Q2^n$ value. (b) SAM value. (c) ERGAS value.

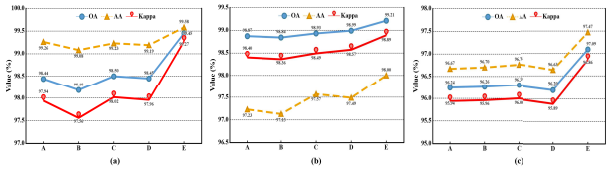


Fig. 16. Quantitative metrics comparison results for the influence of HGFM for HSI classification. (a) PU dataset. (b) PC dataset. (c) HST dataset.

the numbers on the bar chart indicate the quantitative results of category semantics on pansharpening performance, i.e., the quantitative contribution of fine-tuning to the final fusion performance in Stage 3.

4) Influence of Different Scales in MTL-CRMNet: In MTL-CRMNet, the parameter N determines both the number of scales and the network depth. As shown in Fig. 15, increasing N from 1 to 3 enhances multiscale spatial–spectral fusion and improves pansharpening performance, with $N = 3$ achieving the best balance between reconstruction quality and computational efficiency. Beyond $N = 3$, excessive parameters introduce overfitting, leading to performance degradation, most notably when $N = 5$. Thus, $N = 3$ is identified as the optimal tradeoff between accuracy and complexity.

5) Effectiveness of HGFM: We conducted a series of ablation experiments to validate the impact of different HGFM configurations on classification performance across three datasets. In Fig. 16, A, B, C, D, and E represent the following variations: HRHSI in MTL-MGFNet replaced with the result of LRHSI bicubic interpolation, without HGFM, HGFN without the SSEA module, HGFM without Cross-Mamba fusion, and with HGFN, respectively. The experimental results indicate that the classification performance with HGFM surpasses that of the other variations, providing further evidence of the HGFM effectiveness.

6) Influence of Different Receptive Fields in MTL: To verify the impact of local patch size on pansharpening and classification tasks performance, we conducted experiments

TABLE IX

QUANTITATIVE ASSESSMENT OF HYPERSPECTRAL PANSHARPENING AND HYPERSPECTRAL CLASSIFICATION TASK WITH DIFFERENT PATCH SIZES

Dataset	PU	PC	HST
Task / Metric	Hyperspectral Pansharpening		
$Q2^n$	0.9518 / 0.9645	0.9482 / 0.9628	0.9145 / 0.9404
SAM	4.1115 / 3.0609	4.1122 / 3.8876	3.6670 / 3.1796
ERGAS	2.8064 / 2.1529	2.7818 / 2.4689	3.4696 / 2.4251
Task / Metrics	Hyperspectral Classification		
OA(%)	99.15 / 99.45	99.10 / 99.21	96.84 / 97.09
AA(%)	99.40 / 99.58	98.00 / 98.00	97.21 / 97.47
Kappa(%)	98.88 / 99.27	98.73 / 98.89	96.58 / 96.86

TABLE X

AVERAGE QUANTITATIVE ASSESSMENT FOR THE PU DATASET CONSTRUCTED BASED ON MTF

Algorithms	$Q2^n$	SAM	ERGAS
SFIM	0.8459	5.4515	5.5445
GS	0.8317	5.9533	5.4163
CNMF	0.8199	6.2373	5.4549
SFIIN	0.9574	3.3555	2.3973
TransPAN	0.9517	3.4017	2.6285
SSAFF	0.9525	3.4863	2.5099
FPFNet	0.9456	3.9507	3.0873
PANMamba	0.9575	3.3835	2.4893
CMNet	0.9464	3.5772	3.0870
MFT-GAN	0.9167	5.3134	3.8327
MTL-CRMNet	**0.9608**	**3.2214**	**2.2746**

with different local patch sizes. The results in Table IX are that before slash (/), both the classification and pansharpening networks use the same local patch size of 32 × 32, while after the slash, different patch sizes are adopted, 128 × 128 pixels for pansharpening, and 32 × 32 pixels for classification. It can be observed that the fusion performance decreased slightly due to the reduced patch size, while the classification results remained nearly unaffected. The results confirm that performance improvement comes from the proposed bidirectional interaction mechanism, with the task-specific design and mutual guidance, ultimately achieving optimal overall performance.

7) Influence of Different Downsampling Methods on Pansharpening Performance: In this study, a 9 × 9 Gaussian filter was adopted for simulating the degradation process based on three considerations. First, the public datasets lack available modulation transfer function (MTF) parameters, and it is a commonly accepted practice to use Gaussian filtering as an alternative [11], [12]. Second, the MTF of optical systems typically exhibits high-frequency attenuation, which can be effectively approximated by Gaussian filtering in the frequency domain [10], [15]. Third, for fairness and consistency, all compared deep learning-based methods follow Wald's protocol, where Gaussian filtering is conventionally used (see [11], [20], [44]). In order to evaluate the influence of different downsampling methods on the performance of pansharpening, we have also conducted an additional verification experiment using simulated MTF-based physical downsampling. Specifically, following the approach in [59],

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we simulated physically MTF-based downsampling on the PU dataset and evaluated the performance of 11 pansharpening methods, including our proposed MTL-CRMNet. As shown in Tables II and X, traditional methods experienced significant performance degradation on MTF-based data, particularly SFIM and GS methods. In contrast, deep learning-based methods exhibited relatively stable performance across both MTF-based and Gaussian-based simulated datasets. Notably, the proposed method MTL-CRMNet consistently achieved the best results across three objective quality metrics, further demonstrating its robustness and superiority. Based on the mentioned above, it can be concluded that employing Gaussian filtering remains a reasonable and practically effective approach where sensor-specific MTF parameters are unavailable.

V. Conclusion

In this article, we present a unified Mamba-based multitasking framework MambaMTL for HSI pansharpening and classification, thereby facilitating the mutual enhancement between the two tasks. Specifically, the proposed pansharpening subnetwork MTL-CRMNet introduces paralleled MCRLs to refine the reconstructed HRHSI according to scene class semantic information. For HSI classification, MTL-MGFNet is presented to fuse the shared multiscale encoding spatial–spectral features of MTL-CRMNet and the generated HRHSI by the Mamba-based pyramid features fusion strategy for classification. Therefore, the MambaMTL framework effectively combines HSI pansharpening and classification, leveraging their mutual benefits to enhance classification accuracy and produce high-resolution images with rich spatial details. We believe that MambaMTL will provide valuable insights for integrating other important tasks with similar complex interdependencies in various fields. It is worth noting that, although the current pansharpening process employs Gaussian-based downsampling to simulate LRHSIs, future research will focus on integrating sensor-specific MTF to better align with real sensor imaging characteristics. This enhancement will further improve the physical fidelity of the simulation and the practical applicability of the proposed framework.

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